

COMS 3110: Homework 2
Due: October 11th, 11:59pm
Total Points: 50

Late submission policy. An assignment that is submitted one day late will get a penalty of 10%. An assignment that is submitted two days late will get a penalty of 20%. Assignments submitted after two days will not be graded and will get no points. For example, if an assignment is due on Friday by midnight, a submission on Saturday will be penalized by 10%, and a submission on Sunday will be penalized by 20%. A submission on Monday will get no points.

Submission format. Homework solutions will have to be typed. You can use word, LaTeX, or any other type-setting tool to type your solution. Your submission file should be in pdf format. Do **NOT** submit a photocopy of handwritten homework except for diagrams that can be hand-drawn and scanned. We reserve the right **NOT** to grade homework that does not follow the formatting requirements. Name your submission file: `<Your-net-id>-3110-hw2.pdf`. For instance, if your netid is `asterix`, then your submission file will be named `asterix-3110-hw2.pdf`. Each student must hand in their own assignment. If you discussed the homework or solutions with others, a list of collaborators must be included with each submission. Each of the collaborators has to write the solutions in their own words (copies are not allowed).

General Requirements

- When proofs are required, do your best to make them both clear and rigorous. Even when proofs are not required, you should justify your answers and explain your work.
- When asked to present a construction, you should show the correctness of the construction.

Some Useful (in)equalities

- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$
- $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$
- $2^{\log_2 n} = n$, $a^{\log_b n} = n^{\log_b a}$, $n^{n/2} \leq n! \leq n^n$, $\log x^a = a \log x$
- $\log(a \times b) = \log a + \log b$, $\log(a/b) = \log a - \log b$
- $a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}$
- $1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^n} = 2(1 - \frac{1}{2^{n+1}})$
- $1 + 2 + 4 + \dots + 2^n = 2^{n+1} - 1$

1. **(10 pts)** Prefix-Sum Problem: Given an array A of integers, is there a sub-array of A such that if we add all the elements of the sub-array, we get a sum of zero? By sub-array, we mean a contiguous sub-array.

Naive Brute Force Algorithm (Pseudocode):

```
for i = 1 to n {
    for j = i to n {
        sum = A[i] + A[i + 1] + ... + A[j]
        if sum == 0 return true
    }
}
return false
```

Derive the runtime of the above algorithm in detail and determine its Big-O upper bound. You must show the derivation of the end result. Simply stating the final answer without any derivation steps will result in zero points.

2. **(10 pts)** Given an array of integers, write an efficient algorithm to check whether there is a sub-array whose sum is a multiple of a given integer K . You must use a strategy of computing prefix-sums. Give a detailed analysis of the runtime or expected runtime of your algorithm.
3. **(10 pts)** Prove that the height of a heap tree is $\lfloor \log_2(n) \rfloor$ where n is the number of elements in the tree.
4. **(10 pts)**
 - (a) **(5 pts)** Where in a max heap might the smallest element reside, assuming that all elements are distinct?
 - (b) **(5 pts)** Is an array that is sorted in increasing order a min heap? Justify your answer with an explanation.

5. **(10 pts)** Consider an array H of size $n \geq 2$, containing distinct positive integers and an integer K such that $1 \leq K \leq n$. Without sorting the array, write an algorithm with input H and K to find the K^{th} largest integer in H . Your algorithm must use a heap data structure. Write pseudocode for your algorithm and also present an analysis of its correctness and the runtime in terms of K and n .